AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 34 · STUDENT LESSON

Test Diverse Users

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Count the suite. Inspect weak slices. Name untested needs.

Evaluate varied requirements and fictional user conditions without overstating the evidence.

Keep one habit: Recorded outcomes → category rates → separate user-condition traces → priorities → fresh checks.

MAIN LESSON · SECTION 1

Which important failure does the overall score hide?

> Evaluate varied requirements and user conditions without claiming results for untested people.

Your learning goals

I can evaluate across varied cases.

I can report failures with their denominators.

I can separate development **evidence** from final checks.

Recall development and final cases

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- If final cases guide a code change, are they still untouched evaluation cases?

Your end goal

By the end, I can evaluate varied case types and prioritise an important failure slice.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Build the Recommender. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Permissions and Human Review. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Prompt Experiments. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 34: Test Diverse Users

Evaluation must include ordinary, edge, and **underserved** cases.



THINKER MISSION

Run four user profiles; log errors and **fairness concern**s.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Inspect the failed category beside the total

> Designed tests describe a suite, not the frequency of real user needs.

The six-case development record

A fictional chooser passes 5 of 6 development checks but fails its only tie case. An overall **pass rate** can hide that concentrated weakness.

Evaluate ordinary, boundary, invalid, empty and tie cases separately, using requirements rather than duplicated versions of one easy example.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use test diverse users to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

Count six supplied outcomes without overclaiming coverage

> The script counts recorded flags; it does not run the recommender.

Predict and run the complete counting script

```
checks = [{"category": "ordinary", "passed": True},
          {"category": "ordinary", "passed": True},
          {"category": "boundary", "passed": True},
          {"category": "invalid", "passed": True},
          {"category": "empty", "passed": True},
          {"category": "tie", "passed": False}]
passed = 0
for check in checks:
    if check["passed"]:
        passed = passed + 1
print(passed, len(checks))
```

Trace the total and tie slice

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Overall	5 of 6	One failure remains
Tie category	0 of 1	Known weakness
Other categories	Limited cases	No universal guarantee

Read each row from left to right and connect the evidence to the stated rule.

The script reports 5 and 6. About 83.3 percent of these checks pass, but that is not an accuracy estimate for all future use.

The cases were deliberately chosen, their frequencies are artificial, and each category has few cases.

The six development checks include ordinary, boundary, invalid, empty and tie cases. Five pass, but the only tie case fails. Saying five out of six pass is true; saying the tie rule works is false.

Keep both statements visible. A category table helps you choose the next repair. Adding many copies of passing ordinary cases can raise the overall fraction without checking a new condition or fixing the failed requirement.

Explain the deciding evidence

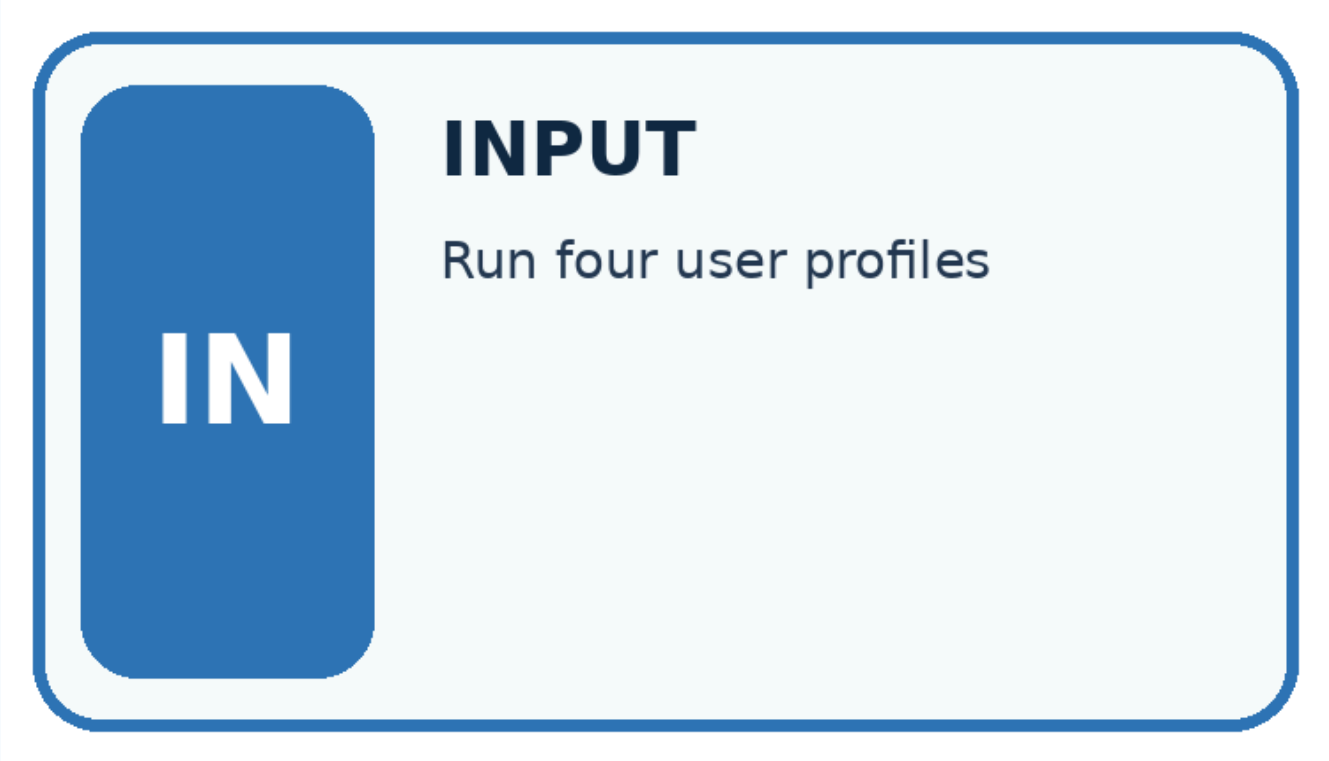
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.



↓


↻
PROCESS
Evaluation must include ordinary, edge, and underserved cases.
↓
OUT
OUTPUT
log errors and fairness concerns.

Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

Look inside the overall result

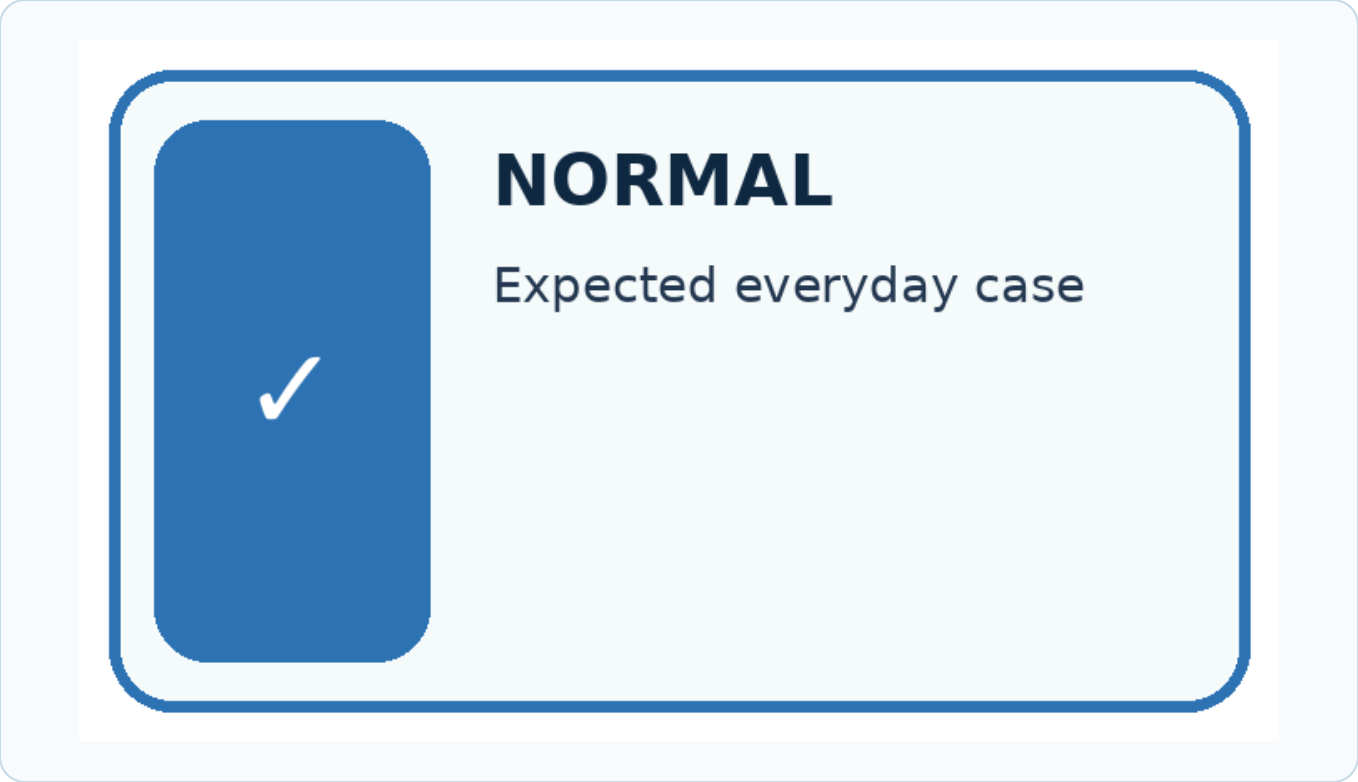
Connect the picture and the explanation

MAIN LESSON · SECTION 4

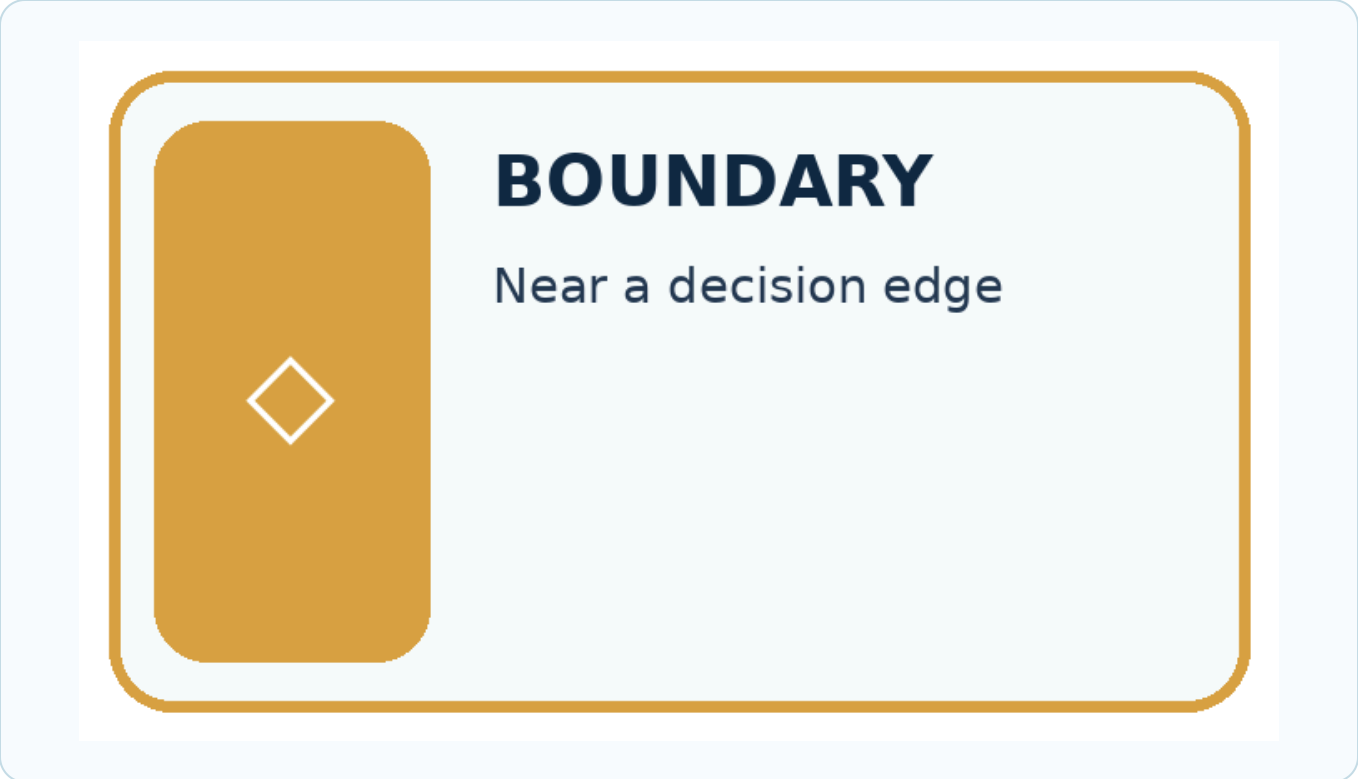
Compare requirement failures and untested conditions

> One comfortable demonstration cannot establish broad coverage.

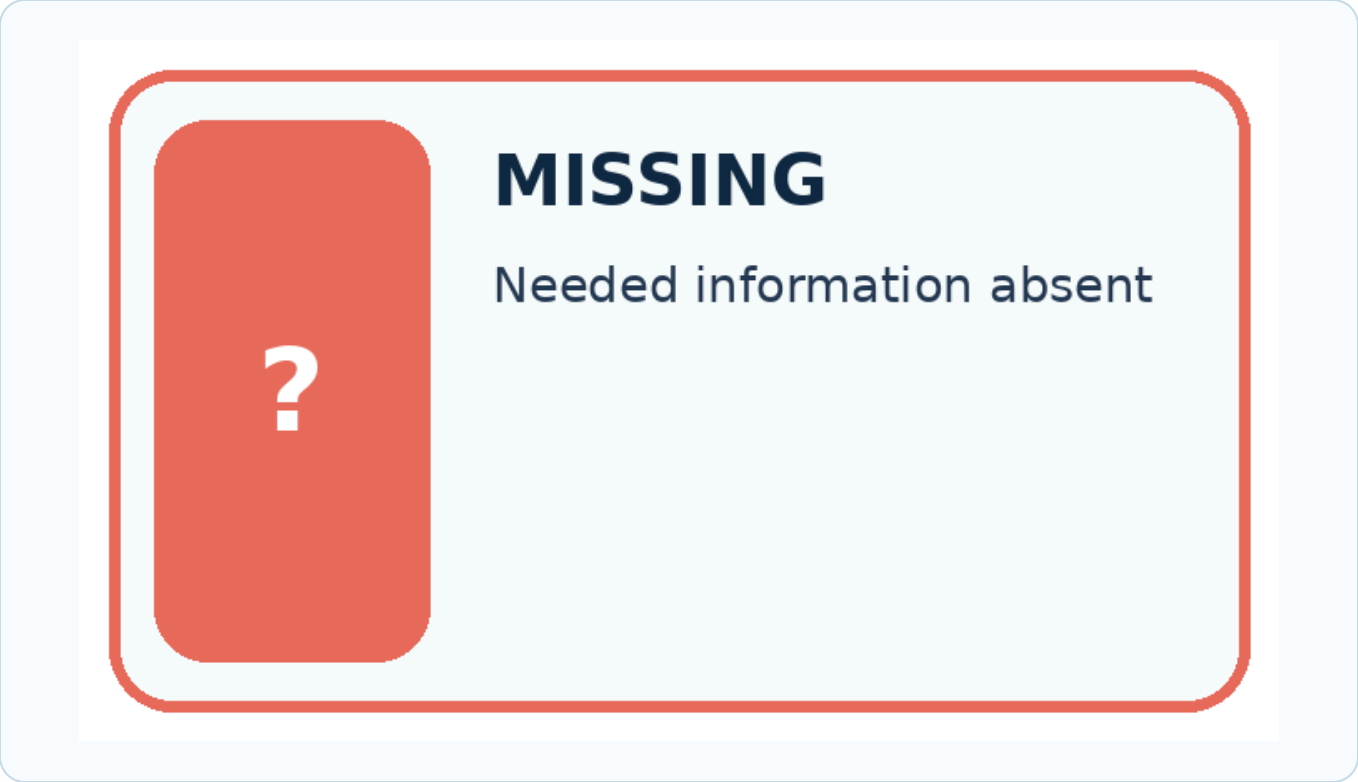
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Someone reports only the ordinary slice.

<details>
<summary>Check after predicting</summary>

Prediction: The claim becomes 4/4 on ordinary cases, not 100% across all tests.

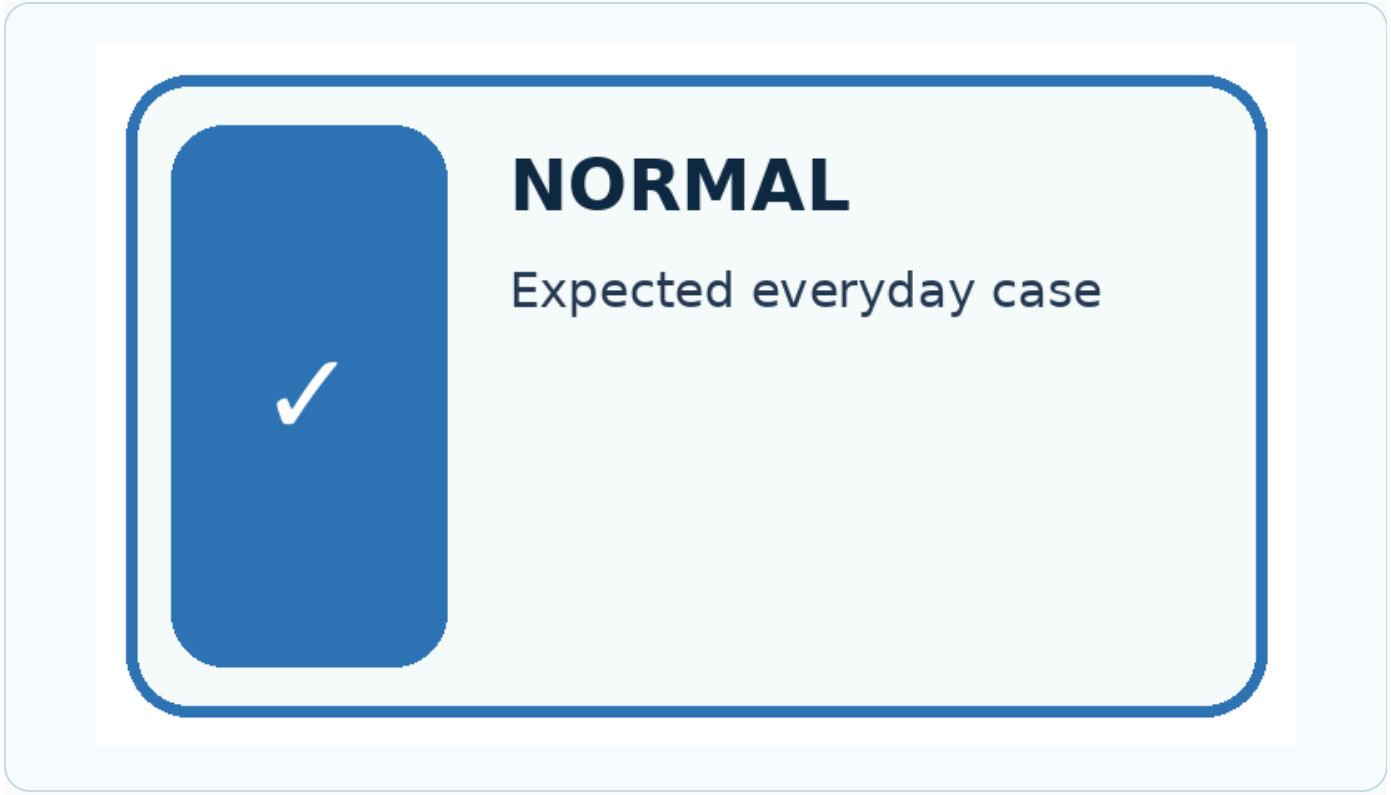
Reason: Eight nonordinary cases and their failures have been omitted.
</details>

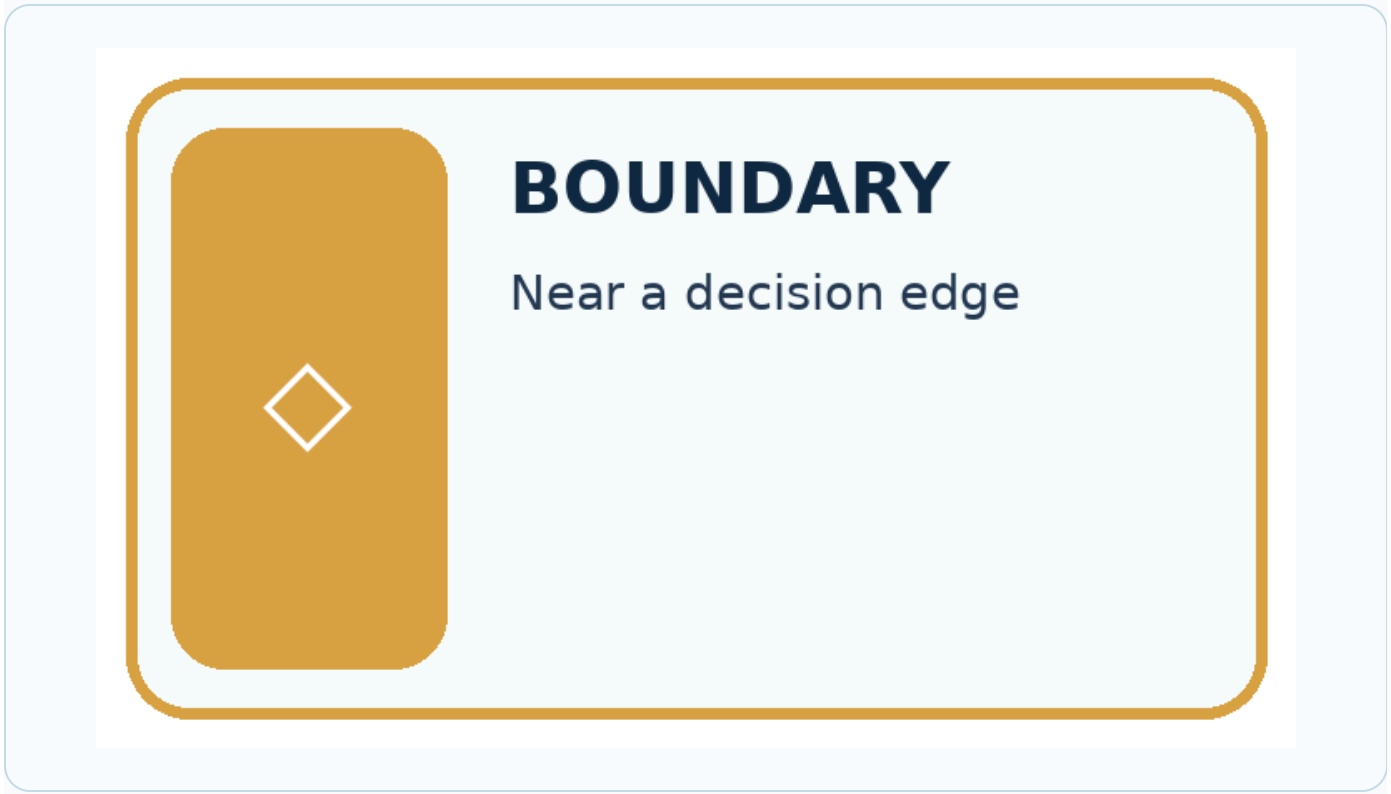
<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

CASE GALLERY

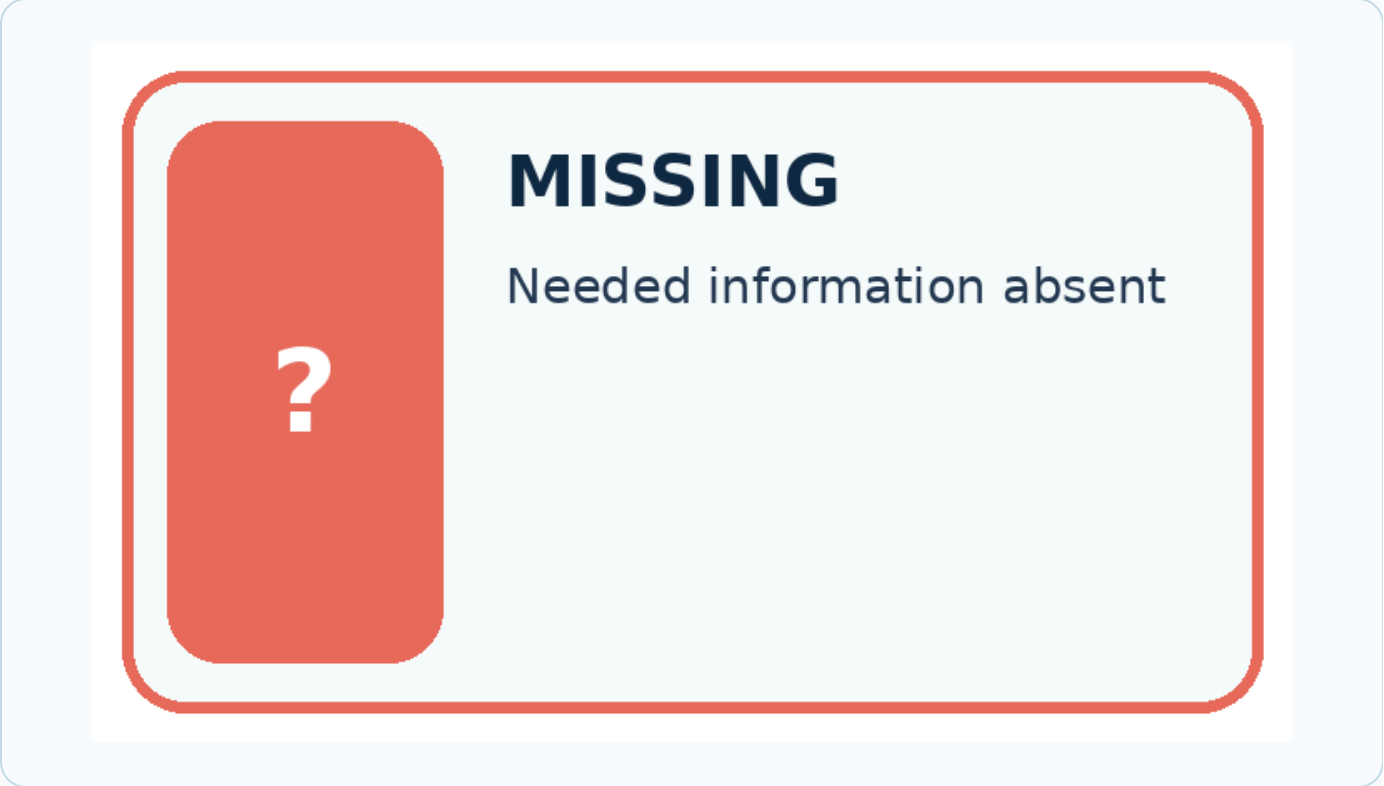
Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.
</details>





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Separate the six-case record from a twelve-case picture

> Keep totals, denominators and condition labels attached to their own datasets.

Predict adding ten duplicate passes

If ten copies of one passing ordinary case are added, what becomes the pass fraction? Does coverage improve?

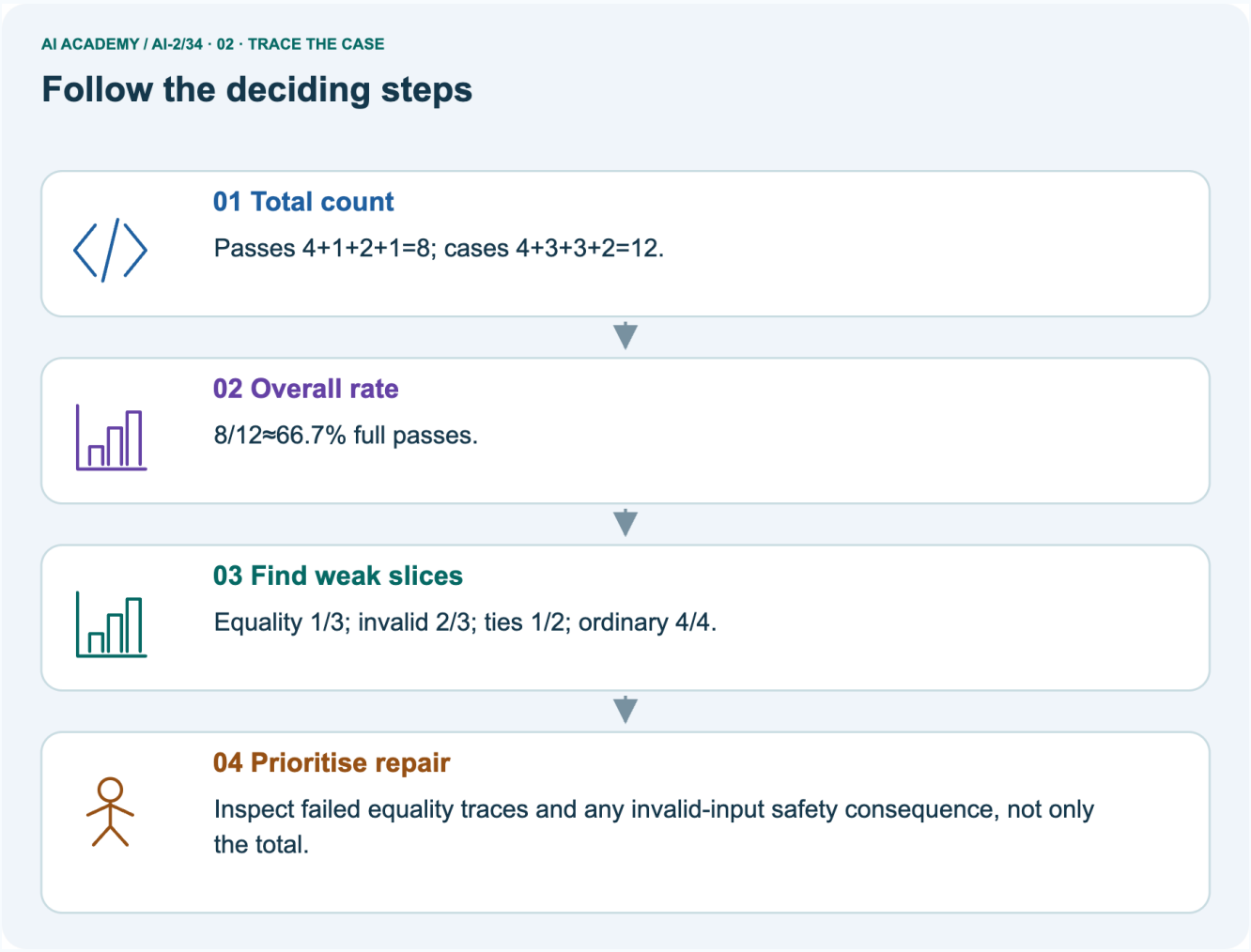
Read the separate twelve-case categories

A fictional recommender is tested on distinct ordinary, equality, invalid-**input** and tie cases.

Recorded full passes: ordinary 4/4; equality 1/3; invalid 2/3; ties 1/2.

Categories are disjoint and every case was run on the same frozen version with the same contract.

Trace overall and per-category results



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Total count	Passes 4+1+2+1=8; cases 4+3+3+2=12.
2. Overall rate	8/12≈66.7% full passes.
3. Find weak slices	Equality 1/3; invalid 2/3; ties 1/2; ordinary 4/4.
4. Prioritise repair	Inspect failed equality traces and any invalid-input safety consequence, not only the total.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

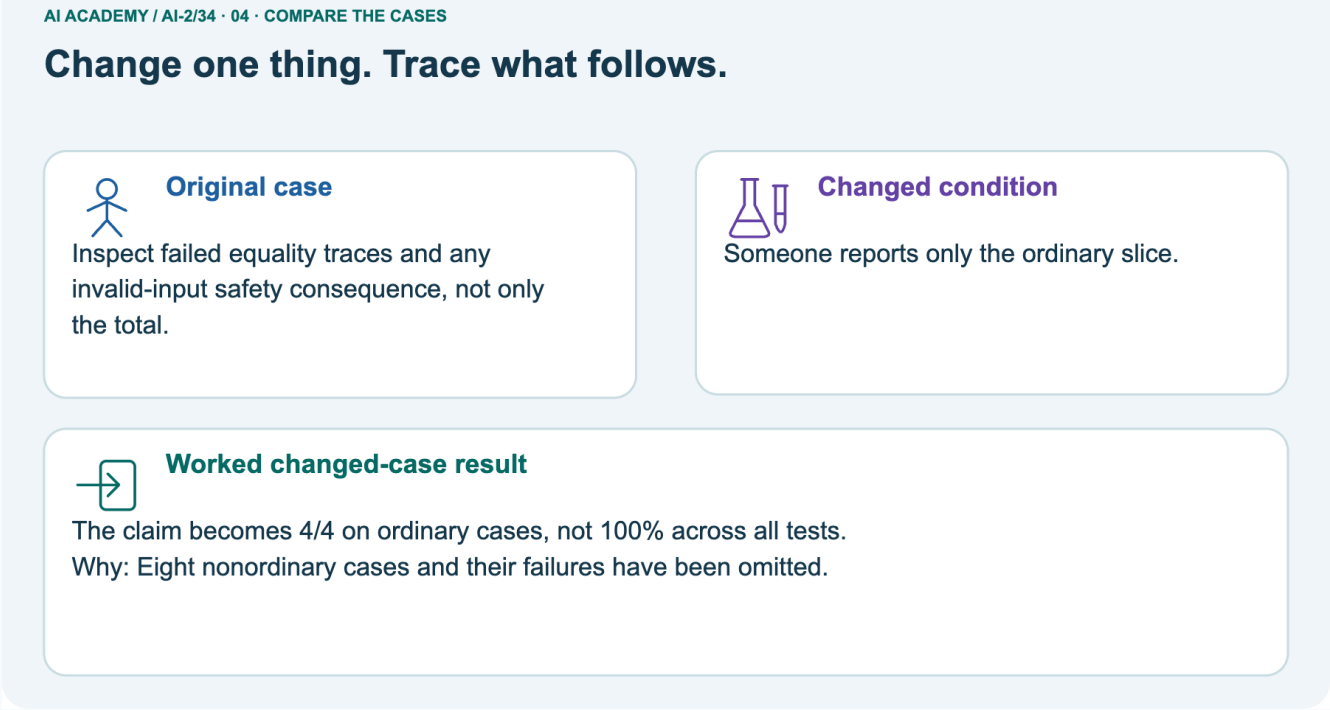
Varied tests expose behavior an ordinary demonstration misses. The overall result combines disjoint cases once each; overlapping labels would require unique-case accounting.

Prioritisation also depends on consequences: a malformed **input** that triggers an unauthorised action can be more serious than a harmless tie-order failure. Retain each failed case and its expected **output**.

Predict reporting only ordinary cases

- Change one thing: Someone reports only the ordinary slice. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the evidence boundary

- Twelve designed cases describe this test suite, not an unbiased estimate of future usage accuracy.
- Read the labelled picture: Can these records establish performance for different user groups?

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 34 • WORKED EVIDENCE

1. Profiles: typical reader; very short-book need; bilingual preference; screen-reader format need.
2. Run the same frozen recommender for all four.
3. Record expected, actual, score evidence, and issue type.
4. Do not average away one group’s repeated failure.

1. Profiles: typical reader; very short-book need; bilingual preference; screen-reader format need.

2. Run the same frozen recommender for all four.

3. Record expected, actual, score **evidence**, and issue type.

4. Do not average away one group’s repeated failure.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

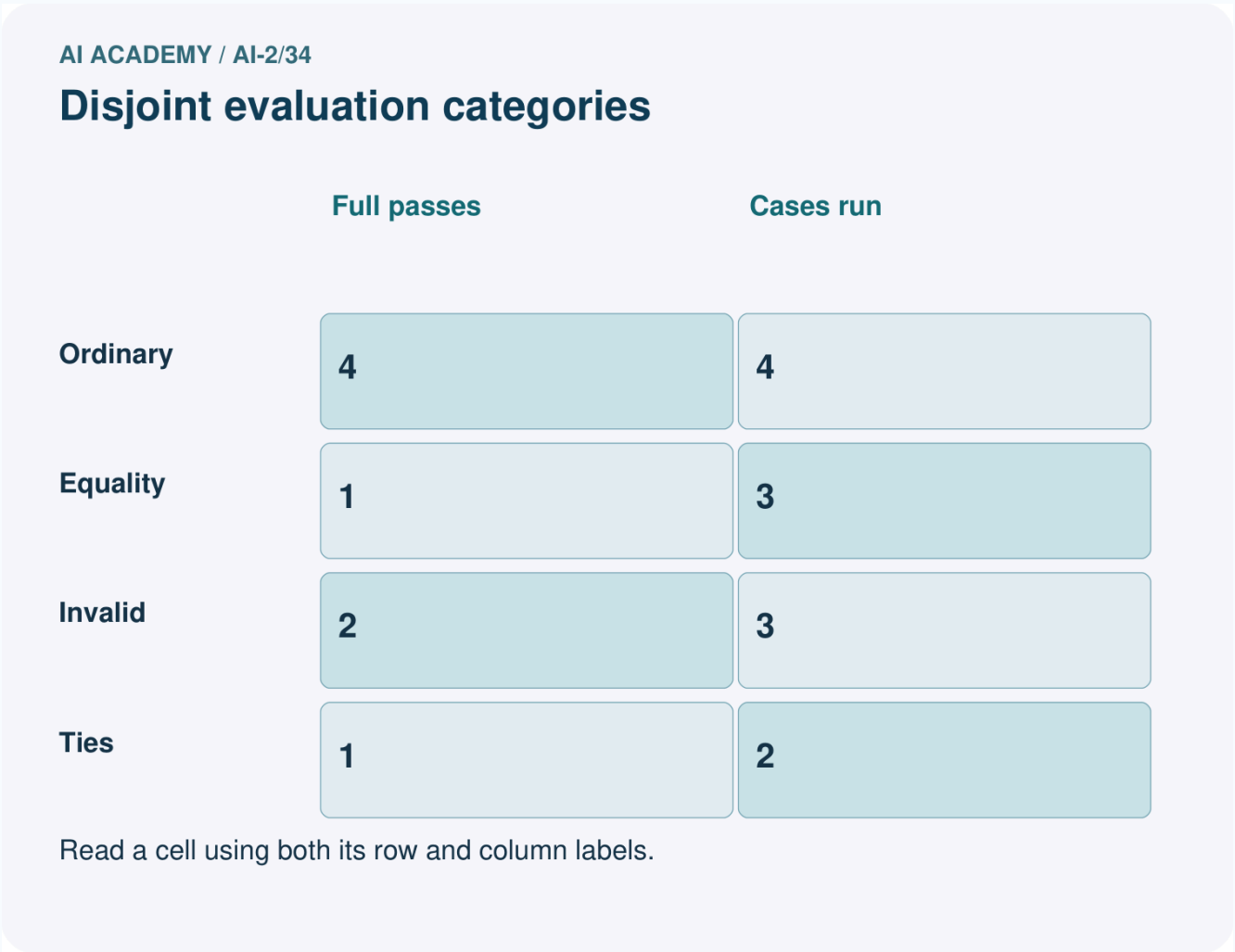
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Test Diverse Users

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/34

Test Diverse Users

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Total count

Passes 4+1+2+1=8; cases 4+3+3+2=12.

2. Overall rate

8/12≈66.7% full passes.

3. Find weak slices

Equality 1/3; invalid 2/3; ties 1/2; ordinary 4/4.

4. Prioritise repair

Inspect failed equality traces and any invalid-input safety consequence, not only the total.

CHANGE ONE THING

Someone reports only the ordinary slice.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Find an untested group claim

> State which conditions the record actually contains.

Repair the coverage argument

Claim to repair: The twelve-case chart proves the recommender works equally well for bilingual readers and screen-reader users. Its labels are ordinary,equality,invalid,ties; no language or assistive-technology result is supplied. Report the actual condition rates, then design separate user-condition checks. Do not relabel failure categories as demographic groups or infer a cause from one failing trace.

Propose the missing check

> **You fix the tie bug using the final tie case. What should the next evaluation report say?**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Good average performance proves every user is served well. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

For a rule-based chooser, evaluate conformance to its policy separately from whether users find suggestions useful. Neither a pass fraction nor an invented satisfaction score proves educational benefit.

Practise a coverage plan with less help

> Keep failed inputs and expected results available for repair.

Retain failures and protect final evidence

Hiding the tie failure in an average conceals an unmet requirement. Preserve the failed **input**, expected result, actual result and version.

If you repair using it, classify it as development/regression **evidence** and reserve new final cases.

Complete main W2

Practise with less help: Plan coverage

Use workbook W2 for the connected example “Evaluating varied cases”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Repair an overclaim before discussion

- Use the worked case for Evaluating varied cases. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Check your reasoning

Explain the pictured coverage boundary

- Can these records establish performance for different user groups?

Explain the picture

My explanation using a pictured step and the given facts

Keep independent evaluation separate

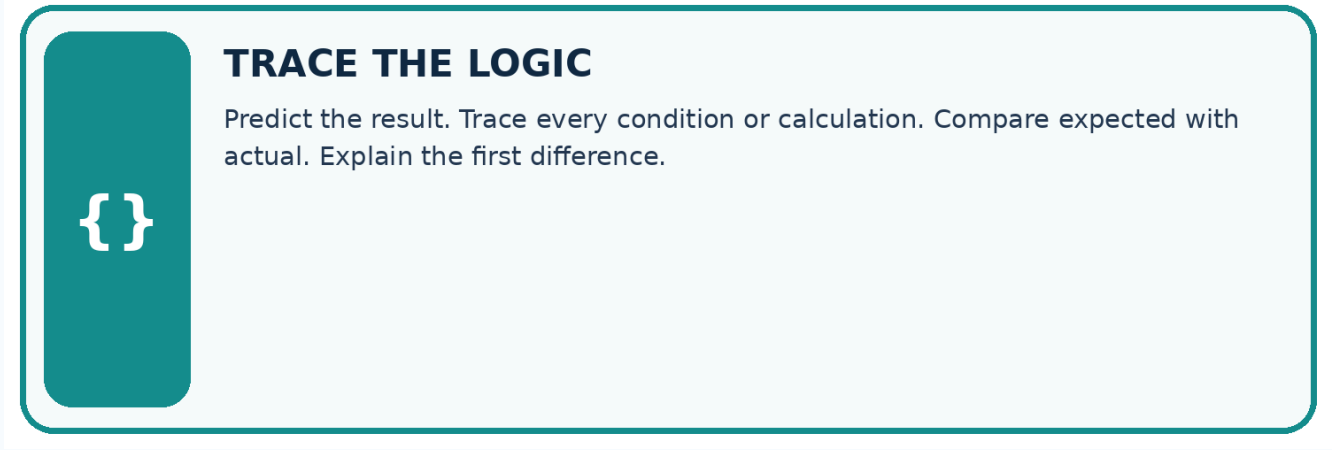
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

MAIN LESSON · SECTION 9

Trace four fully specified fictional user conditions

> Use complete request cards and distinguish contract correctness from unmet needs.

The catalogue and fixed contract

Fictional user-condition investigation: freeze this supplied miniature catalogue and request contract. P:animal,60pages,English,screen-reader-checked text; Q:animal,90pages,Spanish,text with no access check supplied; R:space,40pages,English,image-only. Match requested subject, page maximum and requested language exactly. A request for screen-reader-checked text must have that supplied status; unknown is not a pass.

If none qualify, return no verified eligible option. Tie by ID. These attributes are invented test fixtures, not assessments of real books or users.

The four request cards

Fictional request	Contract fields to test
U1	animal,max100,English,no access-status requirement
U2	animal,max50,English,no access-status requirement
U3	animal,max100,Spanish,no access-status requirement
U4	animal,max100,Spanish,requires screen-reader-checked text

Predict, trace and classify the evidence

1. Predict the eligible IDs for U1–U4 from the complete catalogue. Trace each filter and record a correct no-match outcome as a contract pass, while separately noting the unmet user need or catalogue gap. No live software outputs are supplied: label results a paper contract trace, not a run of an implemented interface. For U4 identify the missing access evidence rather than asserting that Spanish text is inaccessible.

   A real screen-reader claim would require an actual interface, a specified assistive setup and observed checks. Then audit the separate six-case flags and twelve-case counts; do not merge these datasets or call them user-group measurements. Keep the independent eight-case totals for a fresh first attempt.

Keep expected and actual outcomes separate

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

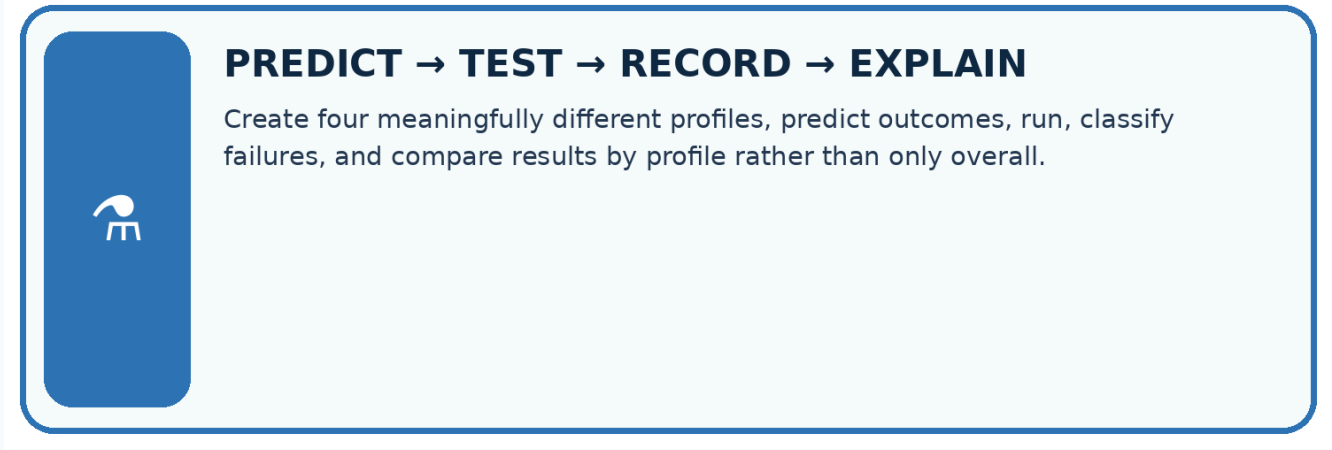
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the supplied fictional P/Q/R catalogue and U1–U4 request cards. Trace fixed constraints, record correct no-match separately from an unmet need, and label the exercise a paper trace. No actual user-group or assistive-technology performance measurement is supplied.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: audit the fresh eight-case suite

> Compute the total and preserve each failed category.

Independent results and counting assumptions

Given ordinary 3/3, boundary 1/2, invalid 2/2 and tie 0/1, compute overall results and prioritize issues.

Assume the listed categories are disjoint and every case was executed once on the same frozen version with the same pass criteria. No case-level failure consequences or user-group results are supplied. Prioritise reviewing both failed categories; state what additional severity evidence would determine the order rather than inventing it.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Evaluate independently

Week 34 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A recommender serves 6 of 6 easy-book requests but 1 of 4 advanced-book requests. Report both results and identify the next test priority.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Use the same test record across ordinary, boundary, missing-data, and access-needs cases.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

The recommender succeeds on 9/10 mystery requests and 2/5 poetry requests. Compute total and separate rates. A team reports only 11/15. Explain what that hides.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep test coverage and user needs visible together

> A passing filter can still reveal an unmet catalogue need.

This week’s concrete contribution

Retain a versioned matrix linking each requirement and user-condition scenario to a complete input, expected output and observed result or labelled unrun plan. Keep correct abstention separate from whether the catalogue meets a request. Add language/access needs only as fictional optional scenarios, not personal profiles. Validate interface access with an actual specified setup before claiming it.

Reused failing cases become regression evidence after repair; reserve fresh final cases.

Build a recommender evaluation matrix with routine, boundary, missing, invalid and tie cases, then add separately planned user-condition **coverage**.

Keep the checkpoint and human choice

THIS WEEK’S CHECKPOINT Use the same test record across ordinary, boundary, missing-data, and access-needs cases.

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

Write or draw your reasoning and evidence.

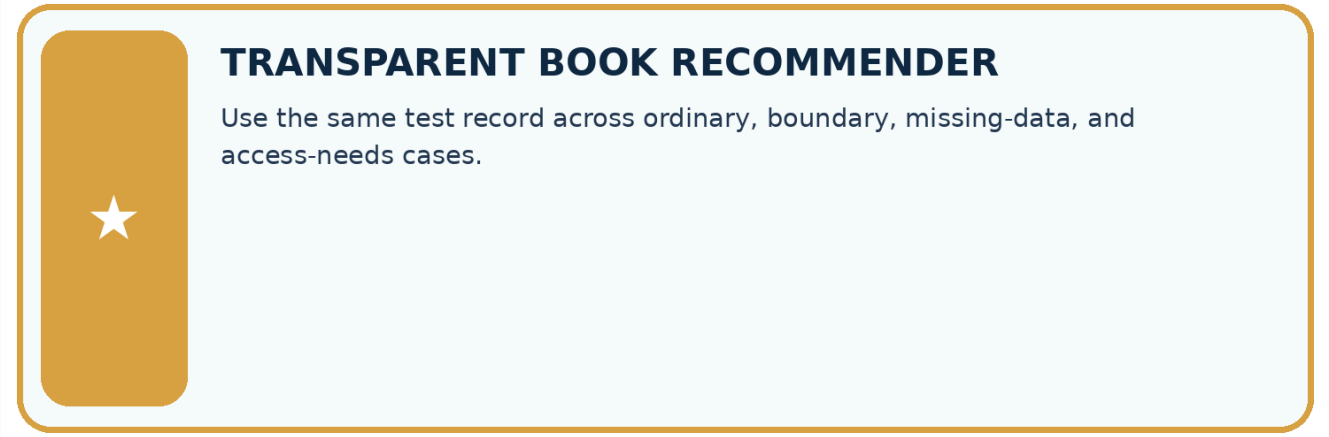
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

Finish: report the total, weak slices and untested conditions

> Do not turn designed-suite results into future-use accuracy.

Explain without notes

- Close your notes. Explain this goal in your own words: Evaluate across varied cases

> **My deciding step, evidence and one remaining question**

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
test profile	A description of the kinds of cases and conditions included in an evaluation.	
edge case	An unusual, boundary or difficult input that can reveal a failure hidden by routine examples.	
underserved	Receiving less adequate support or performance than the intended service requires.	
fairness concern	A reason to examine whether a system creates unjustified differences in treatment or outcomes.	

Terms for the connected explanation

Word	Meaning
Evaluation set	Cases used to assess a specified version

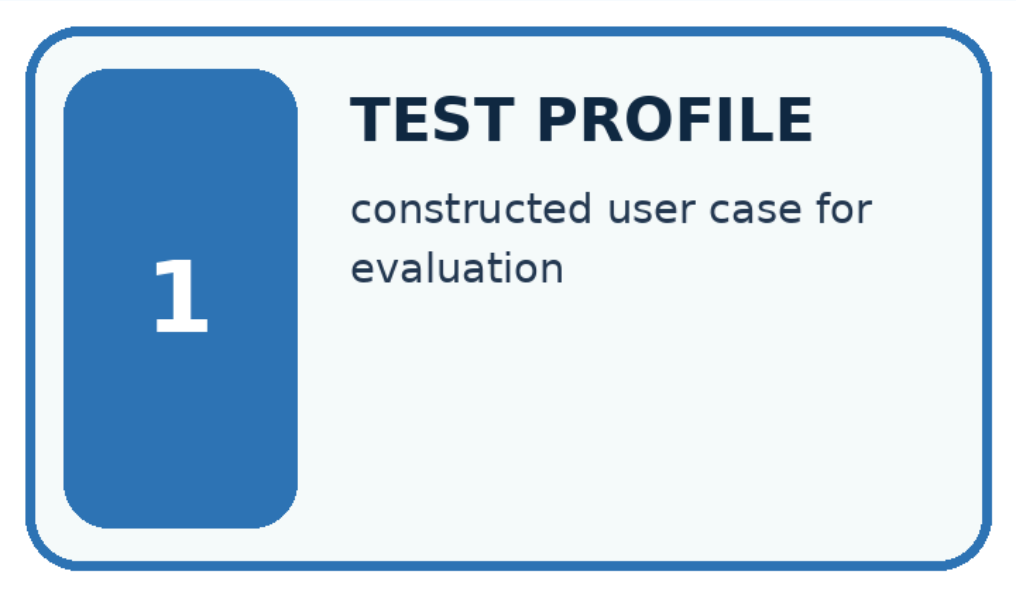
Word	Meaning
Failure category	A type of scenario where an expected result is missed
Pass rate	Passing cases divided by executed cases
Coverage	The variety of requirements and conditions tested

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
EDGE CASE
unusual but plausible input
3
UNDERSERVED
receives weaker outcomes
or coverage



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

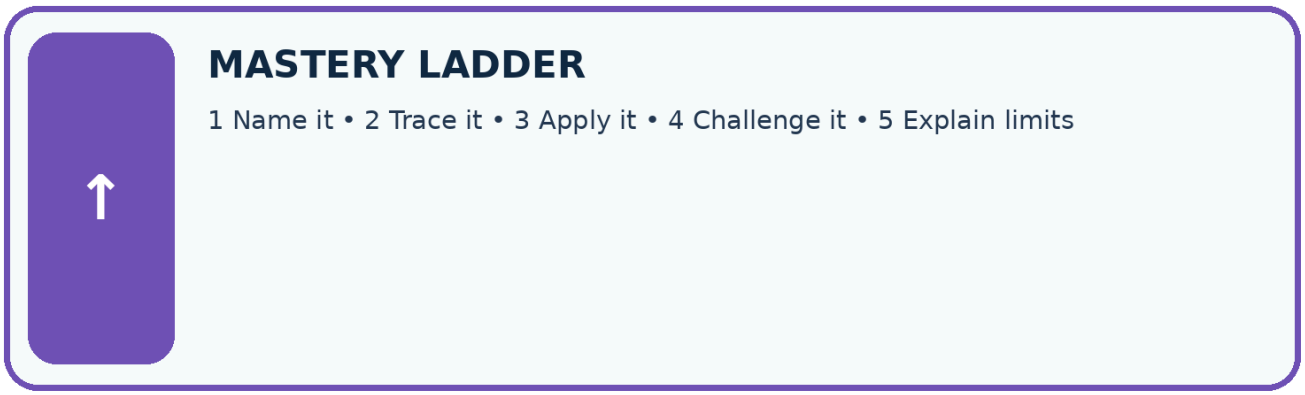
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Audit a new suite independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, **fairness concern**, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

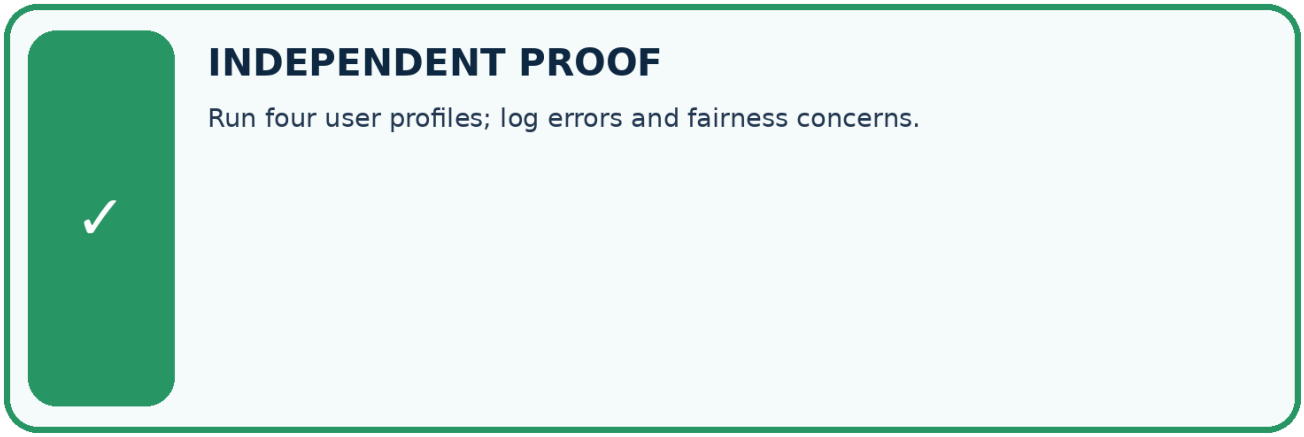
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Audit a new suite independently is the selected independent case.



EXPLANATION CHECK Use disjoint-case denominators, preserve failed categories and state limits on future-use or user-group claims.

Write or draw your reasoning and evidence.

▼ **Extra practice: Compare success for different interests**

REFERENCE · SECTION 13

Extra practice: Compare success for different interests

Try the question before opening its worked explanation.

Read the supplied case and question

A recommender satisfies 5/6 adventure requests and 3/6 poetry requests.

Compare and investigate.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Rates are 5/6 and 3/6. Inspect catalogue **coverage**, labels and scoring; twelve tests do not establish cause.

Evidence or stage	Value or result
Adventure requests	5/6
Poetry requests	3/6

▼ Earlier wording, alternatives and picture transcripts

Week 34 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Compare success for different interests

A recommender satisfies 5/6 adventure requests and 3/6 poetry requests.

Compare and investigate.

Rates are 5/6 and 3/6. Inspect catalogue **coverage**, labels and scoring; twelve tests do not establish cause.

▼ **Reference: original lesson checklist**

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Evaluate across varied cases	Explain the deciding step, show a trace or test, and state one limit.
Report failures with their denominators	Explain the deciding step, show a trace or test, and state one limit.
Separate development evidence from final checks	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional recommender is tested on distinct ordinary, equality, invalid-**input** and tie cases.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional recommender is tested on distinct ordinary, equality, invalid-**input** and tie cases.
- Which labelled step matches this detail? Inspect failed equality traces and any invalid-**input** safety consequence, not only the total.
- 1. Prioritise repair
- 2. Total count
- 3. Find weak slices

- 4. Overall rate

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Prioritise repair Inspect failed equality traces and any invalid-**input** safety consequence, not only the total.

Varied tests expose behavior an ordinary demonstration misses. The overall result combines disjoint cases once each; overlapping labels would require unique-case accounting.

Prioritisation also depends on consequences: a malformed **input** that triggers an unauthorised action can be more serious than a harmless tie-order failure. Retain each failed case and its expected **output**.

Changed-case question

- Changed-case practice: Someone reports only the ordinary slice. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. The claim becomes 4/4 on ordinary cases, not 100% across all tests.

- 3. We can decide the result without examining the changed **input** or the procedure.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The claim becomes 4/4 on ordinary cases, not 100% across all tests. Eight nonordinary cases and their failures have been omitted.

Repair and save your work

Review Build the Recommender (ai-2/33). Goal: Implement a deterministic recommender. Short practice: Explain why B does not win after D is excluded even though it has fewer pages..

Return to this week and explain what you changed.

Review Permissions and Human Review (ai-2/31). Goal: Explain the **mechanism** and terminology of least privilege and tool scopes. Short practice: Which proposal is allowed, and why is an authorised object alone insufficient?.

Return to this week and explain what you changed.

Review Prompt Experiments (ai-2/28). Goal: Explain the **mechanism** and terminology of controlled prompt experiments. Short practice: Name the case P passes and Q fails, and explain why it matters..

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work
</details>

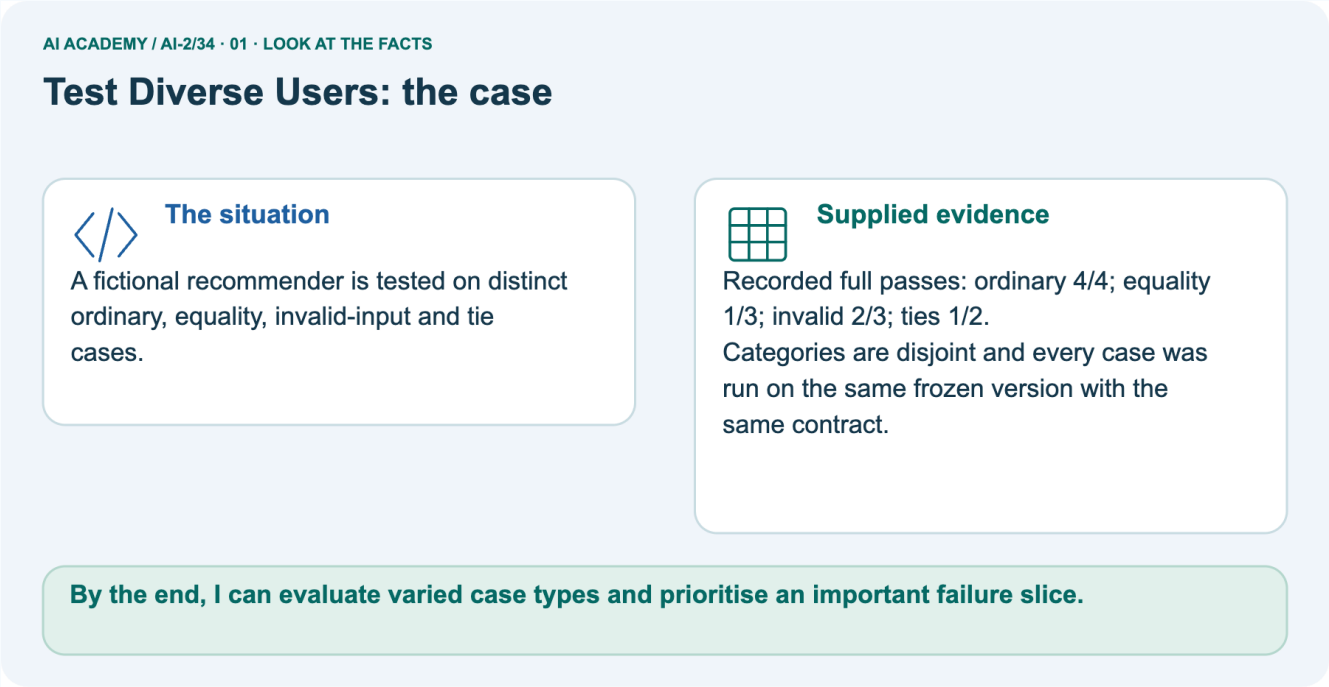
<details>
<summary>Reference: the six-picture journey</summary>
</details>

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Test Diverse Users: the case



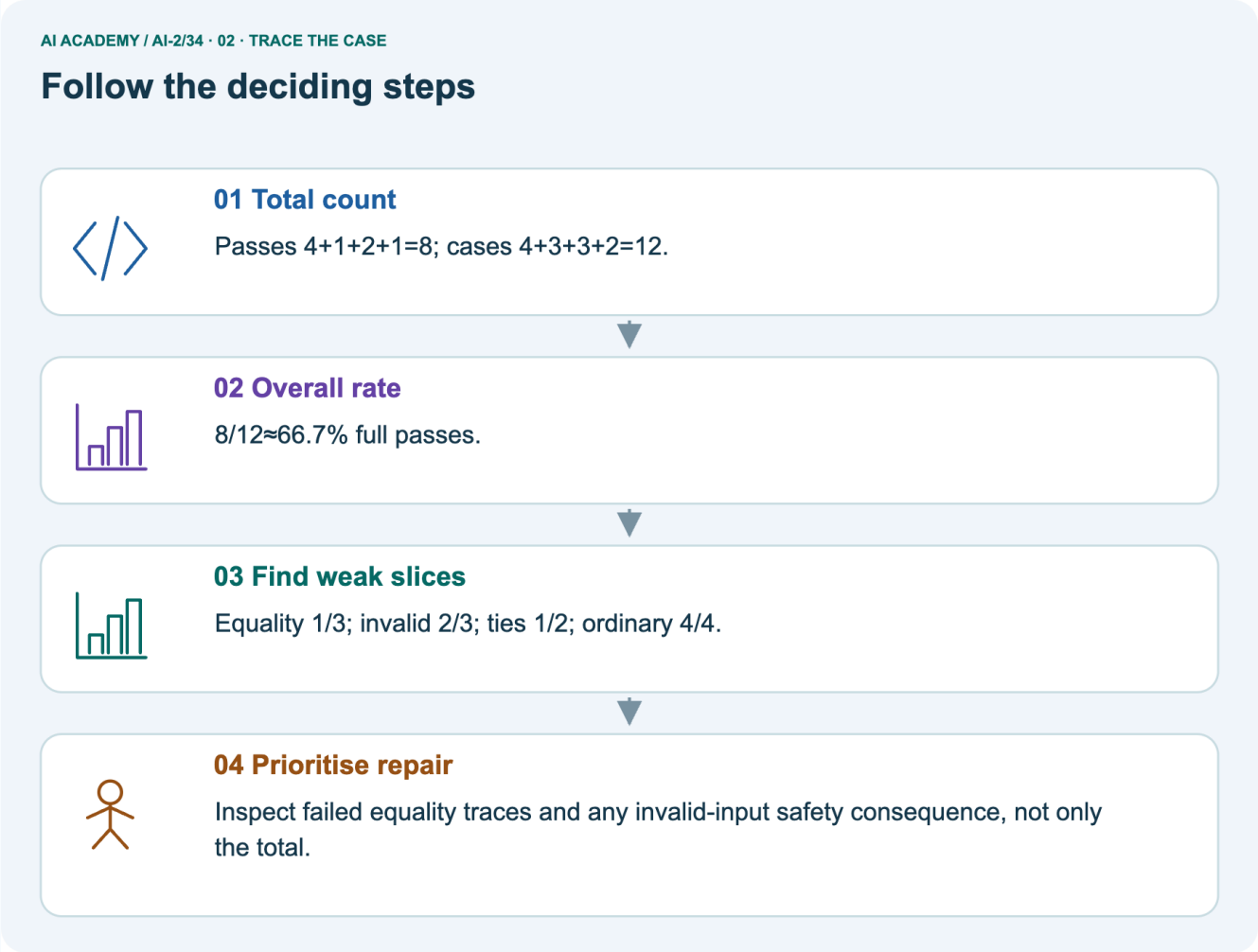
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional recommender is tested on distinct ordinary, equality, invalid-**input** and tie cases.
Supplied facts: Recorded full passes: ordinary 4/4; equality 1/3; invalid 2/3; ties 1/2.

Categories are disjoint and every case was run on the same frozen version with the same contract.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

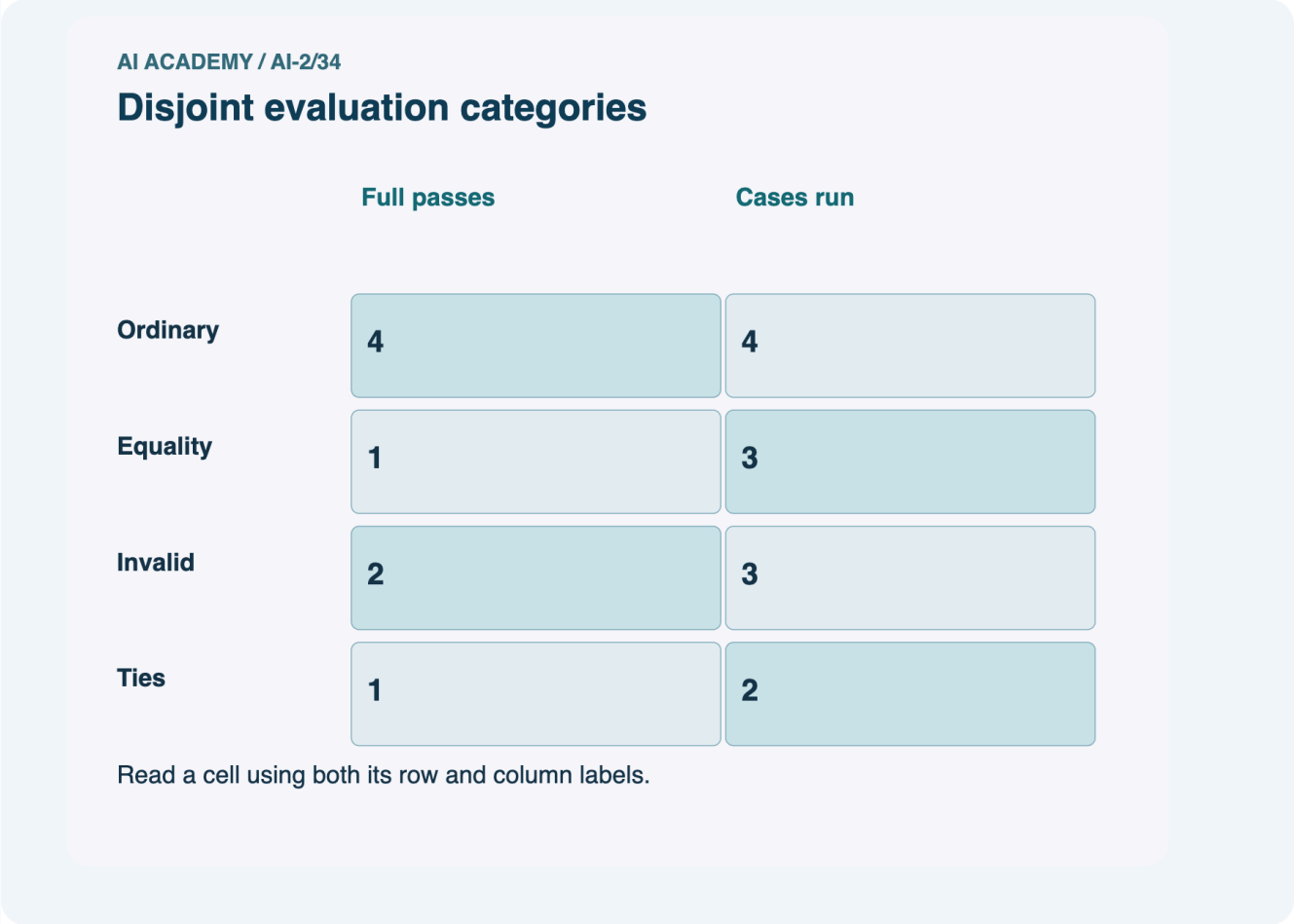


Follow the labelled picture with the matching explanation. Open the vector diagram

Total count: Passes 4+1+2+1=8; cases 4+3+3+2=12. → Overall rate: 8/12≈66.7% full passes. → Find weak slices: Equality 1/3; invalid 2/3; ties 1/2; ordinary 4/4. → Prioritise repair: Inspect failed equality traces and any invalid-**input** safety consequence, not only the total.

Point to each numbered stage and explain which recorded **input** it uses.

03. Disjoint evaluation categories

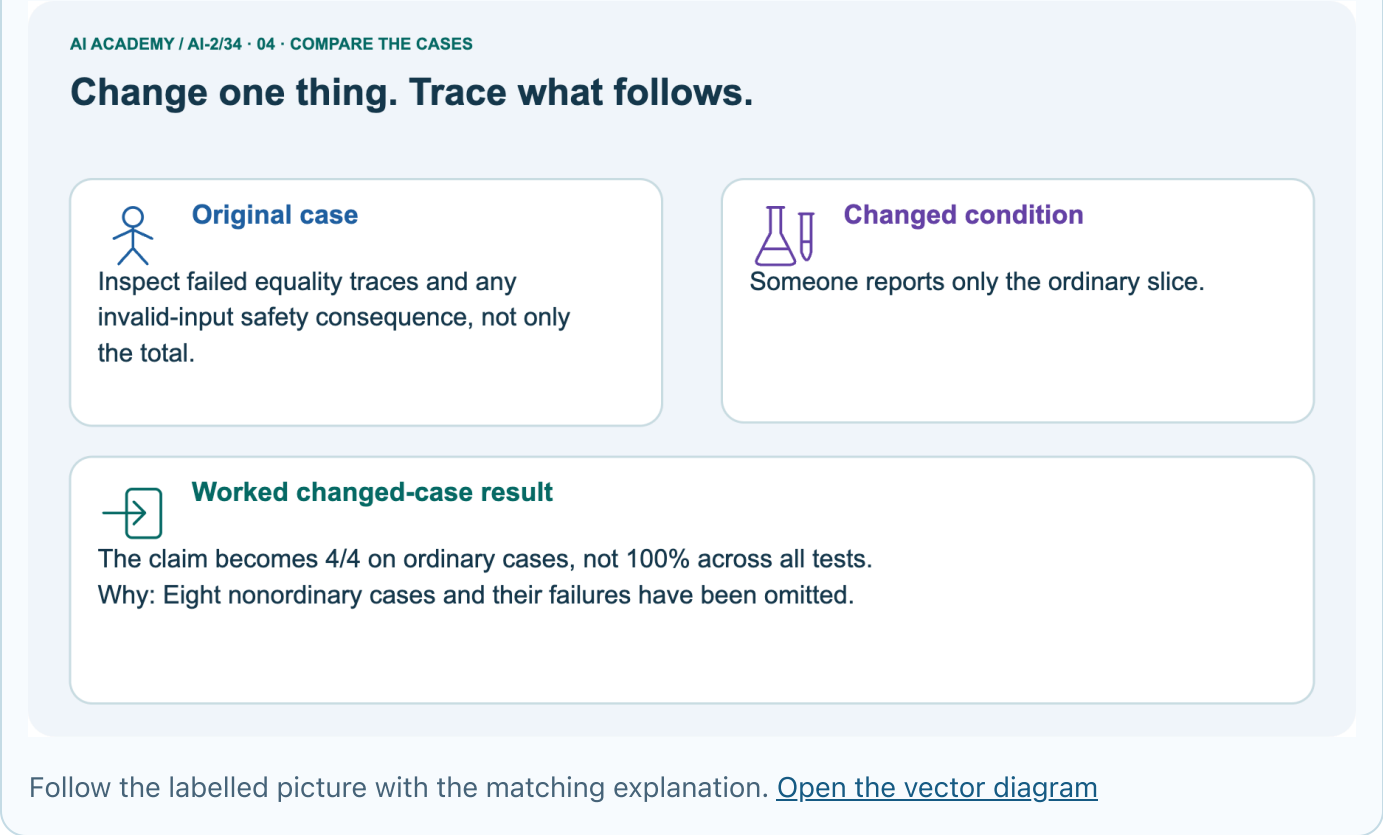


Follow the labelled picture with the matching explanation. Open the vector diagram

Disjoint evaluation categories. Ordinary: Full passes = 4, Cases run = 4. Equality: Full passes = 1, Cases run = 3. Invalid: Full passes = 2, Cases run = 3. Ties: Full passes = 1, Cases run = 2.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



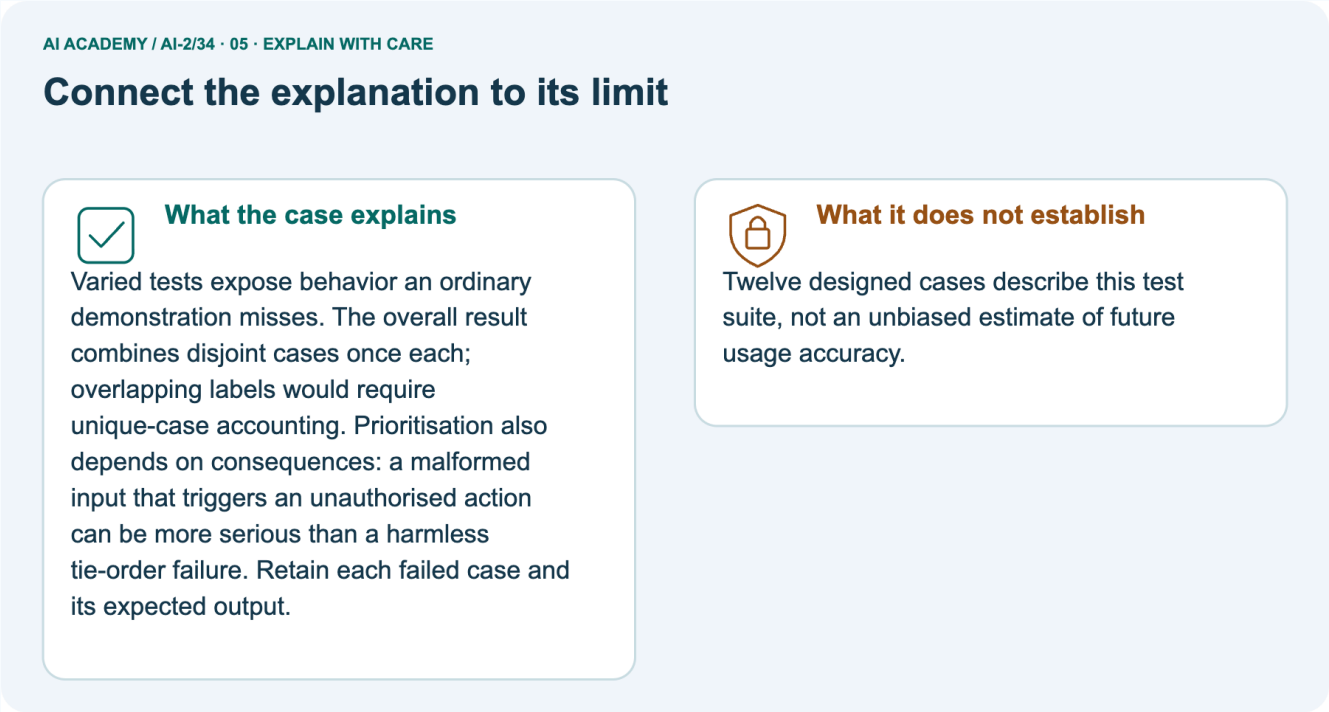
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Inspect failed equality traces and any invalid-**input** safety consequence, not only the total. Changed condition: Someone reports only the ordinary slice.

Expected result: The claim becomes 4/4 on ordinary cases, not 100% across all tests. Reason: Eight nonordinary cases and their failures have been omitted.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

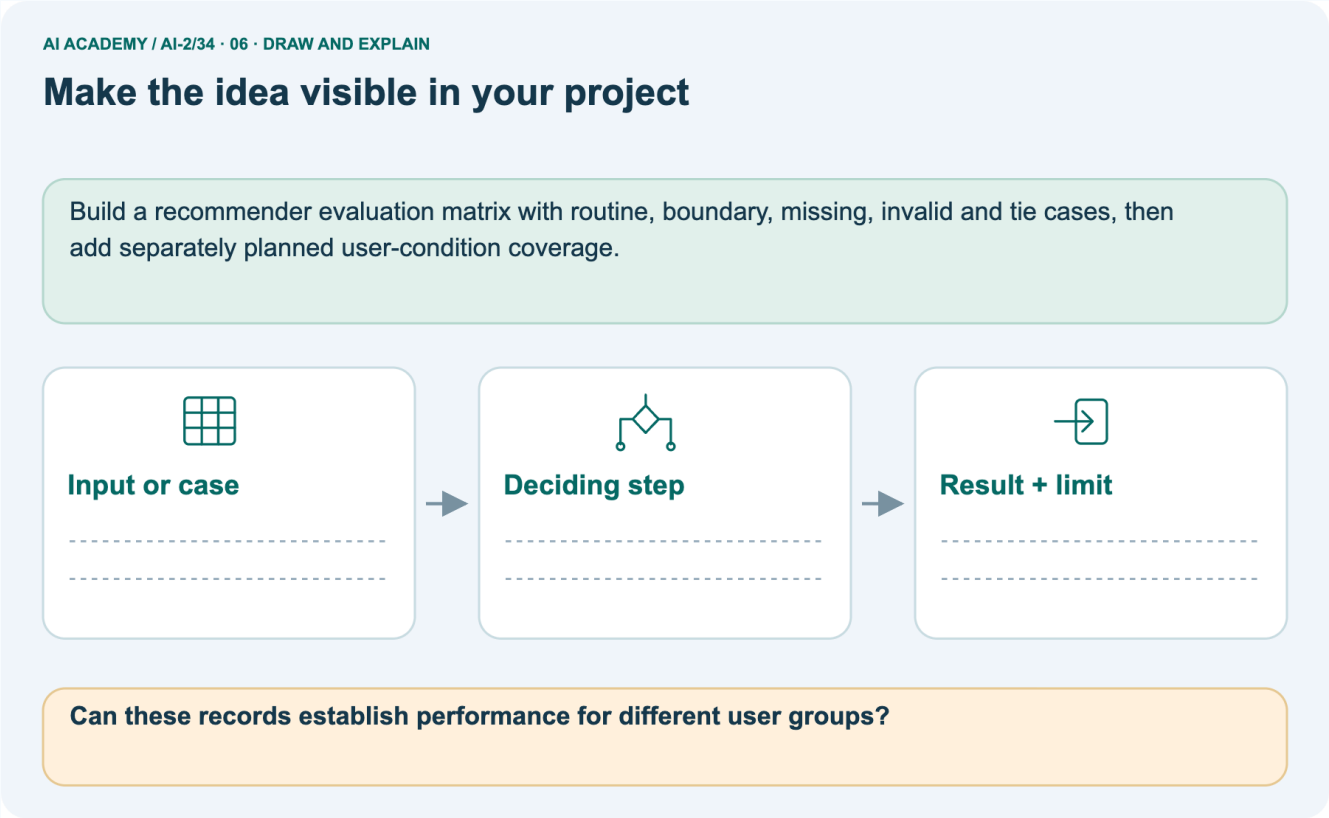
Varied tests expose behavior an ordinary demonstration misses. The overall result combines disjoint cases once each; overlapping labels would require unique-case accounting.

Prioritisation also depends on consequences: a malformed **input** that triggers an unauthorised action can be more serious than a harmless tie-order failure. Retain each failed case and its expected **output**.

Limit: Twelve designed cases describe this test suite, not an unbiased estimate of future usage accuracy.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Build a recommender evaluation matrix with routine, boundary, missing, invalid and tie cases, then add separately planned user-condition **coverage**. Draw your **input**, deciding step, result and limit.

Supported question: Can these records establish performance for different user groups?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Test Diverse Users: the case

02. Follow the deciding steps

03. Disjoint evaluation categories

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.